

1NC--Collapsing the Economy

Hyperscale data centers are the driving factor of US economic growth. Their expansion funds grid infrastructure and accounts for 35-40% of GDP growth.

Fitzgerald '26 [Mark Fitzgerald, CFA, CAIA, Managing Director and Head of Research at Affinius Capital; 6-1-2026; "The US Data Center Market: Powering the US Economy;" AFIRE Summit Issue 21;; <https://www.afire.org/summit/usdatacentermarket/>; 8-1-2026] baybay + mhl

The buildout of **digital infrastructure has become one of the most important drivers of US economic growth.** As **cloud computing and artificial intelligence (AI) adoption continue to expand rapidly across the economy**, investment in data centers, power infrastructure, networking equipment, and related supply chains has accelerated to unprecedented levels. The emergence of generative AI has intensified this trend, as training and operating advanced AI models requires enormous amounts of computational power, driving a sharp increase in demand for hyperscale data center capacity.

Enterprises are increasingly embedding AI into core operations, while consumer adoption has also accelerated rapidly. As a result, **hyperscale cloud providers are undertaking one of the largest capital investment cycles in modern economic history.**

<<FIGURE OMITTED>>

The scale of this investment is now large enough to meaningfully influence broader macroeconomic growth. Beyond direct expenditures on servers and semiconductors, data center development drives substantial downstream investment in power generation, electrical equipment, transmission infrastructure, engineering services, and construction labor. Various economic estimates suggest that AI and data center-related investment **directly accounted for approximately 35%–40% of total US GDP growth in 2025.**

This expansion appears **poised to accelerate further.** As shown in Exhibit 1, capital expenditures by the five largest hyperscale technology companies are projected to increase to approximately \$660 billion in 2026. This compares to approximately \$405 billion in 2025 and represents nearly a tripling of annual capital expenditures in just two years.

These investment decisions are being supported by continued growth in underlying cloud revenues, which have largely kept pace with the rapid increase in capital expenditures, as shown in Exhibit 1. Combined cloud revenues for the largest hyperscale platforms reached an annualized run rate of approximately \$508 billion as of the first quarter of 2026, reflecting the sustained expansion of enterprise cloud adoption and accelerating demand for AI-enabled services. **Hyperscalers continue to characterize the market environment** as supply constrained, with available compute capacity remaining insufficient to satisfy customer demand. For example, Alphabet recently reported that its Google Cloud backlog “nearly doubled quarter-on-quarter to over \$460 billion,” driven by strong demand for enterprise AI offerings and TPU hardware sales.

<<FIGURE OMITTED>>

As a result, demand for data center capacity has continued to accelerate at an unprecedented pace. Net absorption increased from 6.4 GW in 2023 to 9.6 GW in 2024 and reached 16.9 GW in 2025. Momentum has continued into 2026, with first quarter net absorption totaling approximately 9.5 GW, representing the strongest quarterly leasing volume ever recorded in the sector.

While construction activity has accelerated, the market remains fundamentally supply constrained, with demand continuing to outpace the ability to deliver energized capacity across most major markets. Importantly, the primary constraint on new development is access to power. The **US grid** currently supports approximately 36 GW of data center load, yet industry forecasts suggest total demand could approach 80 GW by 2030, requiring a substantial expansion of generation, transmission, and substation infrastructure. Across major US markets, interconnection queues have lengthened materially, with estimated delivery timelines now extending four to seven years in many large metros and as long as five to ten years in Northern Virginia, Columbus, and Northern California.

These power **constraints are reshaping the geography of data center development**. Historically, net absorption was heavily concentrated in a relatively small group of established primary markets benefiting from dense fiber connectivity, large existing ecosystems, and proximity to enterprise demand centers. In 2023, approximately 85% of total US net absorption occurred within primary data center markets. However, as power availability within many core markets has become increasingly constrained, **hyperscale operators and developers have been forced to expand into a broader set of secondary and frontier markets capable of offering large-scale power delivery within commercially acceptable timeframes**. By 2025, over 50% of total net absorption occurred outside traditional primary markets, as shown in Exhibit 2.

Domestic hyperscale data center construction is the sole reason the economy is stable now.

Baggiony-Taylor '25 [Megan Baggiony-Taylor, Deputy Editor for Data Centre Magazine and Telco Magazine, with rich experience in PR, content marketing and media relations. 11-29-2025, "What Risks the US Economy Faces if AI Data Centre Boom Slows," No Publication, <https://datacentremagazine.com/news/how-the-us-economy-faces-risk-if-ai-data-centre-boom-slows>, accessed 7-12-2026]//keeevin + mhl

Heavy **AI infrastructure investment from tech giants is propping up US GDP growth, but any slowdown could expose deep economic vulnerabilities**. AI is no longer just a tool for innovation – it's now a **key component** of the US economy. **From major cloud platforms to chipmakers and hyperscalers, capital is flowing heavily into AI infrastructure**. At the centre of this investment surge are **data centres**. These facilities are now not only essential for powering generative AI, but they **are also holding up the US economy** – and that may be creating a problem. Economists are beginning to question what happens if the AI investment cycle slows. As the traditional growth engines of the economy falter, spending on data centres and compute infrastructure is doing much of the heavy lifting. If that tailwind fades, the entire economic outlook could change. **Data centre spending takes centre stage in GDP growth AI is driving vast capital expenditure, with hyperscale data centres forming the backbone of this surge. Companies including Microsoft, Amazon, Alphabet and Meta** – the four largest spenders – **are expected to invest US\$344bn this year, according to Bank of America. That equals 1.1% of US gross domestic product (GDP) and marks a rise from US\$228bn last year.** Barclays estimates that this capital flow into software, compute equipment and **data centres adds one percentage point to GDP growth in the first half of 2025. With total GDP growth at 1.6% over that period, this suggests that AI infrastructure alone contributes 0.8% – or half the total.** Peter Berezin, Chief Global Strategist at BCA Research The level of dependence is causing concern. **Peter Berezin, Chief Global Strategist at BCA Research, says: "It's certainly plausible that the economy would already be in a recession" without the AI boom.** Strip away this investment and the economic picture looks weaker. Deutsche Bank notes that private business investment outside of AI-related categories has been flat since 2019. Traditional commercial construction – including offices and retail spaces – **beating forecasts AI has become so ingrained in America's economy, that a reversal could risk recession**. Meanwhile, Bank of America projects that the four big tech firms will raise capital expenditures again in 2026 to US\$404bn, though growth in **Economist at Barclays Jonathan Millar, Senior US Economist at Barclays, warns of the impact if markets fall: a 20% to 30% correction in stock prices could cut GDP growth by 1% to 1.5% over the course of a year. A pause in AI investment could trim another 0.5% If it collapses, a full point could be lost.** Peter Berezin summarises the threat: **"If you take a fragile labour market and you kick it with a capex bust, you're probably going to get a recession out of it."** As the **US economy leans more heavily on data centre development and AI infrastructure, the risks of over-dependence grow.** Without continued momentum, the knock-on effects could reach across labour markets, consumer spending, credit and overall growth.

Hyperscalers rely on debt owed to private credit firms who are insanely over leveraged. Debt shocks spread across the entire economy since private credit accounts for 70-80% of hyperscale investment.

Stijn Van Nieuwerburgh 26 [Stijn Van Nieuwerburgh, "Financing the AI Buildout", Columbia Business School, 3-20-2026,
<https://business.columbia.edu/sites/default/files-efs/imce-uploads/svannieuwerburgh/papers/DataCenterJEP.pdf>, accessed 7-24-2026] veer + mhl

The scale of the AI infrastructure buildout raises a natural question: who is financing and owning the underlying assets? While hyperscalers and large technology firms are the primary drivers of demand, they are not, in general, the sole providers of capital. The magnitude of required investment, on the order of trillions of dollars, exceeds what even the largest firms can comfortably fund on balance sheet without affecting leverage, valuation, or strategic flexibility. Historically, major cloud providers tended to own and operate much of their data center infrastructure directly. This model aligned control over physical assets with control over compute workloads and allowed firms to internalize both the costs and benefits of expansion. In the current cycle, however, the scale and speed of investment have led to a more mixed model, in which hyperscalers increasingly combine owned capacity with leased facilities, joint ventures, and partnerships with specialized third-party developers. Figure 1 illustrates the fast growth of capital expenditures on the five main hyperscalers' balance sheets. A central feature enabling this shift from owning to renting is the credit quality of the primary tenants. Hyperscalers such as Amazon, Microsoft, Meta, Google, and Oracle carry investment-grade credit ratings, which are the gold standard in real estate and infrastructure leasing. Long-term contracts with such tenants provide predictable cash flows and make projects attractive to lenders and institutional investors. Figure 1. Hyperscaler Capital Expenditures, Operating Cash Flow, and Capex Intensity Notes: The figure plots capital expenditures and operating cash flow, in dollars, for the five largest hyperscalers using the left axis. The line, plotted against the right axis, shows the ratio of capital expenditures to operating cash flow. The figure illustrates the growing capital intensity of the AI buildout and the extent to which capital expenditures are absorbing an increasing share of internally generated funds. By contrast, many of the firms driving demand for frontier AI, such as OpenAI and Anthropic, do not have a public credit rating and would not, on their own, be able to support large-scale infrastructure financing. In practice, these firms access compute capacity through agreements with hyperscalers, effectively renting their balance sheet strength and creditworthiness. This shift has opened the door to a broader set of capital providers. Data center developers, real estate investment trusts (REITs), and infrastructure funds play a growing role in building and owning facilities, often entering into long-term lease agreements with hyperscalers or other large tenants. Data centers have recently become a core asset class for institutional real estate investors (Green Street, 2026). These arrangements with third-party providers allow technology firms to secure access to capacity without bearing the full upfront capital cost, while providing investors with exposure to long-duration, contract-backed cash flows. Debt financing has expanded alongside this equity participation. Traditional bank lending remains part of the capital structure, but a substantial share of funding now comes from private credit markets and structured finance vehicles. Private credit funds, in particular, have emerged as major providers of capital, attracted by the scale of the opportunity and the perceived stability of contracted revenues. In addition, some projects are financed through securitized structures, in which claims on data center cash flows are packaged and sold to investors. A further extension of this model is the emergence of financing structures backed directly by IT hardware. In addition to shifting data center ownership off balance sheet, hyperscalers and AI firms are beginning to explore ways to finance GPUs and related equipment through asset-backed or lease-based structures.² These arrangements resemble established forms of equipment finance, such as aircraft leasing, in which long-lived assets generate predictable cash flows that support external funding. However, the analogy is imperfect. Compared with aircraft or other traditional collateral, AI hardware has a shorter economic life and faces substantial technological obsolescence risk, as it depends more directly on the evolution of model architectures and demand for compute. As a result, financing structures backed by GPUs may carry greater residual value risk, even as they further expand the scope for distributing AI infrastructure exposure across a broader set of investors.

As a result, a significant portion of the AI infrastructure buildout—spanning both physical facilities and, increasingly, the underlying compute hardware—is financed externally rather than directly by the firms that ultimately use the compute. Estimates by Morgan Stanley Research suggest that more than half of the roughly \$2.9 trillion in investment required to meet hyperscalers' additional compute needs over 2025–2028 will be funded by outside capital (Morgan Stanley

Research, 2025). Across the full investment, Morgan Stanley projects an approximate 60–40 split between equity and debt, with hyperscalers providing a substantial share of the equity capital alongside third-party private equity investors. Within the debt component, **private credit is expected to account for the majority (about \$800 billion, or roughly 70 percent)**, with the remainder coming from corporate debt (approximately \$200 billion, 17 percent) and structured finance (about \$150 billion, 13 percent). **The aggregate 60–40 equity-debt split, however, understates the degree of leverage at the asset level.** Much of the internal equity provided by hyperscalers is used to finance IT equipment, while a large share of external debt is tied to the construction of data centers and associated power infrastructure. **As a result, leverage on these physical assets is likely to be significantly higher, often in the range of 70 to 80 percent**, consistent with standard project finance and real estate lending practices. In this sense, **the shift toward external, asset-backed financing may increase effective leverage relative to a model in which infrastructure is financed directly on hyperscalers' balance sheets through unsecured corporate debt.** The resulting structure has several important economic implications. First, ownership of AI infrastructure is increasingly dispersed across a broad set of investors, allowing the sector to tap into deeper pools of capital and potentially improving diversification of the funding base. Second, this expansion is accompanied by a growing complexity in the number and structure of claims on underlying cash flows, as equity holders, private credit funds, banks, and structured finance vehicles each hold distinct contractual positions. Third, the allocation of financing across 2Recent credit ratings agency research points to growing investor demand for AI-linked collateral, including both infrastructure and equipment-backed exposures (Moody's Ratings, 2025). 8 asset classes, where IT equipment is largely financed with internal equity while real estate and power infrastructure are **financed with external asset-backed debt, can lead to higher effective leverage at the asset level than suggested by hyperscaler balance-sheet leverage measures.**

Taken together, these developments separate the users of compute from the owners of physical assets and from the ultimate bearers of financial risk. Hyperscalers and AI firms contract for capacity through leases and service agreements, while a diverse set of investors holds long-duration claims backed by infrastructure, real estate, and IT hardware assets. Understanding how this separation reshapes the distribution of risk across investors and institutions is the focus of the next section. The financial structure that has emerged around AI infrastructure separates the users of compute from the owners of physical assets and from the ultimate bearers of financial risk. Hyperscalers and AI firms contract for capacity, while a wide range of investors, through equity stakes, private credit, and structured finance vehicles, hold claims on the underlying cash flows. This separation enables the rapid scaling of infrastructure by drawing on a broad pool of capital. At the same time, it reshapes how risk is distributed, priced, and monitored within the financial system. Importantly, this redistribution of risk does not imply a reduction in aggregate risk. Instead, it changes who holds the risk and how transparent those exposures are. The resulting system bears similarities to other forms of securitized and infrastructure finance, in which long-duration assets are funded through layered claims held by diverse investors. Understanding these mechanisms is essential for assessing both the financial capacity for a further AI buildout and its potential financial vulnerabilities. A useful illustration of these dynamics is provided by the recently announced Hypertion data center financing. The project involves the development of approximately 2.5 GW of capacity, with a total investment of around \$20 billion, making it one of the largest single data center developments to date. Meta was initially the sole owner of the project but subsequently sold an 80 percent equity stake to the private credit firm Blue Owl for approximately \$2.5 billion. The resulting joint venture, named Belgnet, then raised \$27 billion in external debt in October 2025. The debt was rated A+ by Standard & Poor's, one notch below Meta's own corporate credit rating, making it the largest individual investment-grade corporate debt issuance in U.S. history. Importantly, the capital structure implies a leverage ratio of roughly 50 percent debt, computed as \$27 billion in debt out of \$50 billion in asset value, far above what would typically be observed on the balance sheet of an investment-grade corporate bond issuer. At the same time, the structure exhibits a high degree of financial engineering. The project is organized through a network of bankruptcy-remote special purpose vehicles (SPVs), illustrated in Figure 2, which issue debt backed by contractual claims on future lease payments. This structure increases the capacity of the system by making it more difficult to trace ultimate exposures. The lease contract itself reflects a tension between the needs of capital providers and tenants. While lenders require long-duration, stable cash flows to support large-scale financing, tenants value flexibility in a rapidly evolving technological environment. In this case, the solution takes the form of a sequence of shorter-term leases, supplemented by a residual value guarantee that replicates most aspects of a longer-term commitment. Specifically, Meta is the sole tenant of the Hypertion data center and has committed to five successive four-year leases, beginning in 2025 when the facility is the book value of equity and 4% when using the market value of equity. 10 expected to become operational—and extending through 2045, when the Belgnet bond matures. At each renewal date, Meta retains the option to terminate the lease for one or more portions of the campus. If it exercises this option, the underlying asset is offered for sale, and Meta must cover any shortfall between the realized sale value and a contractually specified guaranteed minimum value. This residual-value guarantee (RVG) protects lenders by ensuring that the debt can be repaid even if the tenant exits. The RVG naturally has counterparty risk. The arrangement reflects a trade-off between flexibility and financing. Shorter-term leases provide the tenant with optionality in a rapidly evolving technological environment, while the RVG restores the long-duration, predictable cash flows required by creditors. At the same time, the accounting treatment of these commitments creates a form of regulatory arbitrage. Under current GAAP rules, Meta recognizes neither future lease obligations nor the RVG on its balance sheet, even though one of these two liabilities is certain to materialize. As a result, Meta's reported leverage understates the economic commitments embedded in the lease contract. This understatement will only be corrected at the time that either the lease is renewed or the RVG option is exercised. Moody's Ratings (2024b) finds that hyperscalers have about \$970 billion in combined lease commitments, \$460 billion of which are not reflected on balance sheet. This off-balance sheet debt arrangement is not without cost. Belgnet's debt was issued at a yield of 6.58 percent, at least 100 basis points higher than what Meta would likely have paid to issue an equivalent amount of unsecured corporate debt. Over the life of the bond, this difference translates in over \$5 billion of additional interest expense. This higher cost of debt is ultimately passed through to Meta in the form of higher lease payments. The benefit to Meta lies in preserving an asset-light balance sheet. By financing the project off balance sheet, the firm avoids adding substantial capital expenditures and debt that could put pressure on its credit rating and, more importantly, on its equity valuation. Meta is valued more like a high-growth software company than a capital-intensive infrastructure provider, with correspondingly higher valuation multiples. Maintaining that valuation generates gains that far outweigh the higher cost of debt associated with external, asset-backed financing. The Hypertion-Meta transaction stands as a landmark deal that may well serve as a template for the next generation of data center finance. The transaction highlights how the separation of ownership, control, and risk-bearing can give rise to forms of regulatory and financial arbitrage. By locating assets and liabilities within off-balance sheet entities, firms can obtain large amounts of secured financing while preserving balance sheet flexibility and credit ratings. At the same time, the resulting structure increases leverage at the asset level and distributes exposure across a wide range of investors, raising questions about how risks are measured and monitored within the system.

The rapid expansion of AI infrastructure is supported by a financial structure that redistributes risk across a wide set of investors and institutions. **While this structure facilitates investment at scale, it also introduces new forms of risk that are shaped by the technological, contractual, and financial characteristics of the sector. A first source of risk arises from concentration and counterparty exposure. Many large data center projects are effectively single-tenant facilities, with hyperscalers serving as the primary or sole occupants.** While these tenants currently have strong credit profiles, **their financial positions are tied to the evolution of the AI ecosystem, including demand for AI services and the profitability of downstream applications. In addition, hyperscalers are themselves exposed to a relatively small number of large AI model developers and enterprise clients. This creates the possibility of correlated risks, in which a downturn in AI demand propagates through tenants to infrastructure investors.** Given that hyperscalers frequently own part of their data center footprint, a contraction in compute demand may prompt a reallocation of activity from leased facilities to owned capacity. A second risk relates to technological obsolescence. The rapid pace of innovation in AI hardware and system design implies that data centers built for current generations of chips may become less competitive over time. Increases in power density, cooling requirements, and interconnect technologies can render existing facilities costly to retrofit or upgrade. The threat is not limited to incremental hardware upgrades. More disruptive shifts, such as quantum computing for select computational workloads or edge computing that moves processing onto phones, vehicles, factories, and other local devices, could structurally weaken demand for large centralized data center campuses. As a result, assets that are financed over long horizons may face shorter effective economic lives than anticipated, creating a mismatch between asset durability and financial structure. A third set of risks stems from power and infrastructure constraints. The viability of large AI facilities depends critically on access to reliable and scalable electricity, as well as on permitting and transmission infrastructure. Delays in grid interconnection, rising electricity costs, or regulatory opposition at the local level can affect both the timing and profitability of projects. These constraints introduce an additional layer of uncertainty that is largely outside the control of developers and tenants. A fourth source of risk arises from potential bottlenecks in the supply chain for AI hardware. The expansion of compute capacity depends critically on the availability of advanced semiconductors and related components, including GPUs, high-bandwidth memory, and specialized networking equipment. These, in turn, rely on a highly concentrated and technologically complex global production chain, spanning chip design, fabrication, and the manufacture of advanced lithography equipment. Capacity constraints at any point in this supply chain can slow the pace of deployment. Geopolitical risks further amplify these concerns, as key segments of production are geographically concentrated, for example, the chip manufacturer

TSMC in Taiwan. These supply-side constraints raise the possibility that the projected expansion of tens of gigawatts of AI infrastructure may not fully materialize. While such an outcome could increase the value of existing facilities by limiting future competition, it would pose risks for investors whose returns depend on the timely deployment and utilization of new capacity. In particular, debt holders in projects collateralized by data center leasing revenues may face shortfalls if the required hardware cannot be delivered at scale, causing actual compute capacity to fall below contracted or underwritten levels. Financial structure introduces a fifth dimension of risk. As discussed in the previous section, the separation of ownership and use is often accompanied by high levels of asset-level leverage and by complex contractual arrangements. **While these features allow more capital to flow into the sector, they can also amplify losses in adverse scenarios.** These dynamics echo a broader literature on securitization, which emphasizes both its ability to expand credit supply and distribute risk, as well as the incentive and information problems it can create. DeMarzo (2005) shows how securitization can improve risk sharing but may weaken monitoring incentives, while Coval, Jurek and Stafford (2009) highlight how structured finance can obscure tail risks and amplify losses when underlying assumptions fail. Lenders who originate construction loans or mortgages for data centers with the intention of selling the loans to structured finance vehicles may lower underwriting discipline. For early evidence in this direction, Gete and Mercader (2026) argue that an originate-to-distribute model is emerging in datacenter mortgage lending. **The combination of high leverage, long-duration debt, and uncertain future demand creates the potential for valuation adjustments that are transmitted across multiple layers of investors.** More broadly, the interaction of these **risks may give rise to systemic effects.** The AI ecosystem is characterized by increasing circularity, with interlocking strategic relationships along the value chain linking chip designers, cloud providers, AI model developers, and data center operators. These interdependencies create channels through which **shocks to one segment, for example, a decline in AI demand or financial stress among model developers, can propagate across the entire system.** Recent asset pricing evidence is consistent with increasing systematic risk. The estimated stock market beta of publicly-traded data center REITs has risen from roughly 0.5 in earlier periods to around 1 in recent years, as shown in the black line in Figure 3, indicating that these assets now co-move closely with the broader equity market. This shift suggests that data centers no longer behave as relatively defensive infrastructure investments, but instead are increasingly exposed to the same macroeconomic and technological risks that affect large technology firms. Given the central role of these firms in equity markets, the risks associated with AI infrastructure may therefore be more systemically important than what traditional real estate or infrastructure investors may be accustomed to. Data center stocks also increasingly comove with the non-AI component of the overall stock market. This undermines the case for data centers as a good hedge for AI-related risk. Over the past three years, the exposure of data centers to non-AI stocks has become as high as its exposure to AI stocks.

None of these risks imply that adverse outcomes are imminent. A more optimistic view emphasizes the strength of underlying demand for AI compute, the continued scaling of model capabilities, and the strategic importance of AI infrastructure to both private firms and governments. Under this scenario, high utilization rates, sustained growth in AI applications, and ongoing technological improvements support strong and stable cash flows for infrastructure assets. In addition, the involvement of well-capitalized hyperscalers and institutional investors may enhance the resilience of the system. At the same time, the scale and structure of the current buildout make it important to understand how risks could materialize under less favorable conditions. Historical experience with large infrastructure booms, from railroads to telecom, shows that periods of rapid, capital-intensive expansion accompanied by growing leverage are often followed by financial stress when demand falls short of expectations or financing conditions tighten. The interaction of technological uncertainty, financial leverage, and complex contractual arrangements in the AI sector suggests that similar dynamics could emerge. Even if such outcomes do not materialize, careful consideration of their probabilities is essential for the adequate pricing of risk and for evaluating the long-term sustainability of the AI infrastructure boom.

Investors and private credit firms won't get paid back until the projects are completed

Davies '26 [Paul J. Davies; 7-13-2026; "AI's \$5.8 Trillion Buildout Needs Every Bond Flavor It Can Sell"; Bloomberg;

<https://www.bloomberg.com/opinion/articles/2026-07-14/ai-s-5-8-trillion-buildout-needs-every-bond-flavor-it-can-sell>] veer + mhl

The **data-center building boom has sparked an explosion of debt funding** even though it has barely begun. But the investors financing this engine of the artificial-intelligence revolution aren't paying enough attention to the terms of their lending. Some of the deals being underwritten offer proper protections against construction delays or problem tenants; others, much less so. Bondholders need to get a grip on the detail in the documents before things start to go awry.

The five hyperscalers – Alphabet Inc., Amazon.com Inc., Meta Platforms Inc., Microsoft Corp. and Oracle Corp. – are expected to invest \$5.8 trillion in the computing capacity they need to run AI by 2030, according to estimates from Goldman Sachs Group Inc. All apart from Oracle have very healthy cash flows, but that income and more will be ploughed into chips, servers and the vast sheds in which they're housed.

The big five have already raised billions of dollars from corporate bond sales in the past 18 months, including nearly \$200 billion just this year. At the same time, several have used off-balance-sheet joint ventures to fund individual data centers with about \$90 billion raised against more than a dozen projects.

These standalone JV deals will remain an important source of money for the AI buildout, because the industry's funding needs are so vast it will have to tap as many kinds of debt as it can. Despite the complexity and variety of terms in these deals, bond buyers have focused more on the attractions of backing the big tech companies and the AI trend rather than scrutinizing the details of each project. That's a choice they could come to regret.

Take the nearly \$11 billion of bonds tied to three data centers for cloud-operating startup FluidStack Ltd., backstopped by Alphabet's Google division. Each project has raised dedicated funding via privately placed bonds for structures that will be leased by FluidStack when completed. The underwriting shortcut is to rely on the strength of Alphabet as guarantor — but some of the terms are very different.

One has a full guarantee from Google to assume the lease if the cloud operator fails, which is great for the lenders. Another has a guarantee from Google but at a 25% rental discount, according to analysts at Citigroup Inc. The third has a formula for the rent that Google would pay linked to operating costs. The backstops for the latter two should cover the debt costs, but they offer less comfort than the first.

There are further differences. One of these deals has a guaranteed maximum price of construction agreed with the engineers building it, a clause that's missing from the other two, which might be less good for the health of the project. Another variation: One project has a clause that lets FluidStack and Google walk away if construction is delayed by more than six months, which would be terrible for bondholders.

There are more examples of friendly and unfriendly lender terms in these deals and others that are critical to judging the risks of a default. Only one of the FluidStack deals got an investment grade credit score from ratings companies — the \$3.25 billion HUT 8 DC LLC project in St Francisville, Louisiana, which has a full lease guarantee from Google, a maximum construction price and no walk-away rights for the tenant.

But the borrowing costs of most data center JVs are similar despite the wide variety of terms, and the bond values move closely in line with each other, suggesting little investor discrimination between them. Investors have treated the asset class as homogenous, according to credit analysts at Goldman Sachs. They suspect yields will diverge more over time as the protections — or lack thereof — become more important.

The key risk that investors probably underappreciate is completion risk, according to Anubhav Arora, a senior director at Fitch Ratings Inc. "They're looking at the company providing the lease guarantee, the Google or Meta, and not thinking about the fact that rent doesn't start getting paid until the project is completed," he told me.

Lenders won't get their money back until and unless this cash starts flowing. Fitch has given its highest credit rating so far in this area to a project in Michigan ultimately backed by Oracle. The software giant may be financially weakest of the hyperscalers, but on this project it has guaranteed to start paying rent by a certain date whether the data center is built or not.

The demand for these facilities is so strong that developers are starting to run up against real-world constraints, such as securing enough labor to build them. Arora says this could give landlords a bit more power over tenants and their hyperscaler backers, which could be good for future lenders. But it also means deals are likely to remain idiosyncratic, rather than the market converging on standard terms. Everyone needs to keep studying the documentation closely.

Given this complexity, why don't Google and its peers just issue cheaper, simpler corporate bonds to fund this massive infrastructure bet? Apart from Oracle, most have very little debt on their books and could afford to borrow hundreds of billions each in this way.

Pullback on AI capital expenditure from hyperscalers creates shocks that make it impossible to service debt. This collapses credit markets and banks.

BIS 26 [Pablo Hernández de Cos, Optimism Around, "I. Progress and peril", Annual Economic Report 2026, Bank for International Settlements, 6-27-2026, <https://www.bis.org/publ/arpdf/ar2026e1.htm>, accessed 7-24-2026] veer + mhl

AI has the potential to raise productivity significantly over the coming decade. Task-level studies consistently report large efficiency gains, often to the tune of between 20 and 50% in time savings (Graph 10.A). Aggregate productivity growth estimates tend to be more conservative at less than 1% over a long horizon, reflecting challenges in adopting the technology at scale and integrating it with production processes. Still, there are further upside productivity gains, particularly if the technology improves to the point at which knowledge creation can be automated. The potential implications of such transformative AI for growth, income distribution and monetary policy are profound (Box 3).

Artificial intelligence (AI) has the potential to differ fundamentally from earlier waves of technological progress. Previous general purpose technologies, such as the steam engine, electrification and information technology, raised workers' productivity by providing them with better tools. AI could go further by augmenting the production of knowledge itself. If, at some point, AI systems can improve their own capabilities and "create" technology and ideas, the macroeconomic consequences could be profoundly different from past innovations. A key constraint on long-run growth, namely the rate at which humans can generate new ideas, could be lifted.

Recent research has begun to formalise this possibility. Trammell and Korinek (2023) argue that if AI capital becomes a sufficiently close substitute for human labour, the economy can transition from the regime of constant exponential growth to one of accelerating super-exponential growth. As machines autonomously improve themselves, the economy acquires a self-reinforcing engine of growth. Jones and Tonetti (2026) identify a supply side counterforce. If different economic tasks are strong complements, the overall output would be constrained by the task that is improving slowest, the weakest link. AI progress in automated tasks could then fail to translate into faster aggregate growth if essential human performed tasks hold output back. Trammell and Korinek (2023) also argue that economies currently operate in this complementarity regime, implying that the transformative scenario remains a possibility rather than a present reality.

These frameworks are supply side oriented – they focus on the productive potential of AI and assume that consumer demand keeps pace with supply. But as AI advances, automation increasingly diverts income from labour, which is spent on goods and services, into further AI investment. The consumer base could erode as productive capacity expands. Forward-looking firms, recognising the shrinking future market for AI-produced goods and services, may find it unprofitable to invest in innovating and automating the next task. Productivity stalls not because of technological limitation, but because the demand to justify further capacity expansion is missing. The demand bottleneck becomes the binding constraint.

A unified framework that nests the supply side predictions as special cases, but incorporates demand bottleneck, synthesises the range of possibilities and provides insights for monetary policy. The framework features a task-based production structure in which AI capital can automate a rising share of tasks. It has two classes of households (capital owners and workers) to track who bears the costs of adjustment and features a forward-looking free-entry condition governing firms' decisions to invest in further innovation and automation. In calibrating the model, trend productivity, markups, product turnover and longevity of innovation rents are set to standard empirical benchmarks. The model economy's labour share and r -star before the introduction of transformative AI are matched to data and established estimates. By varying the scope of automation and innovation's reliance on consumer demand, the model generates scenarios that include: (i) business as usual; (ii) bounded productivity gains; and (iii) explosive growth. Critically, it identifies a fourth possibility, the demand bottleneck.

The projections for aggregate output highlight the wide uncertainty surrounding AI's growth impact (Graph C1). Under the business as usual scenario in which labour remains the binding input, output continues to grow at its historical trend rate of 2% annually. A bounded productivity boost would shift trend growth permanently by a constant margin – which has not occurred since the Industrial Revolution. Under the transformation AI scenario, in which AI becomes a self-reinforcing growth engine, output growth expands exponentially and the labour share falls towards zero. Under the demand bottleneck scenario, output growth rises before falling below its historical trend in the long run as automation stalls. Since every displaced worker is also a lost consumer, the spending that rewards innovation eventually shrinks, leading to a slowdown. This scenario produces a labour share decline that is less severe than the transformative case, but worse than the other two cases.

The analysis also sheds light on the implications for the natural rate of interest, or r -star. Under the business as usual scenario, r -star stays at its pre-AI level. Under the bounded productivity boost and transformative AI scenarios, higher productivity growth raises the marginal product of capital and r -star increases. Under the demand bottleneck scenario, r -star initially rises while supply side forces dominate but then falls below its pre-AI baseline as the bottleneck takes hold. The r -star implications thus depend on parameters that are uncertain today: how much AI profits and innovation rely on consumer demand, how quickly competition within the AI sector erodes margins, and how fast AI products and infrastructure become obsolete. The corresponding medium-term inflation pressures move with r -star in direction, turning disinflationary if the demand bottleneck dominates and inflationary if expected productivity lifts demand on net.

The transition to a more productive AI-driven economy entails risks, however. As more capable AI tools find applications in more tasks and occupations, labour displacement could intensify. Whether or not AI advances create new jobs – or expand demand for existing ones – sufficiently to make up for such displacements remains uncertain. Unlike past general purpose technologies, AI competes directly with human cognitive abilities, possibly narrowing the scope for workers to move up the value chain or find new non-disrupted tasks. To date, such disruptive labour displacements have yet to occur at scale. But there are signs of possible adjustments to come. In earnings calls, more firms are acknowledging potential productivity gains from AI, signalling their intent to automate an increasing share of production processes and engage in labour substitution (Graph 10.B). Consistent with this, US sectors with higher exposure to AI have also seen higher productivity gains, partly at the expense of lower employment growth relative to other sectors (Graph 10.C).

In the near term, the ongoing AI investment boom raises questions about the sustainability of the current economic expansion. The five largest hyperscalers are set to spend over a trillion US dollars on AI-related capital expenditure from 2025 through 2026. These commitments are outpacing earnings and the free cash flow of these firms, leading some to issue debt to raise additional financing (Graph 11.A). This investment race may be partly driven by the perception that only a small number of players with superior technology will ultimately dominate the market shares. The intense competition raises the risk of firms over-committing resources to investment projects with still uncertain returns, leaving all firms vulnerable to disappointments in AI payoffs. Model analysis based on such contest motives highlights the downside risk of current AI exuberance. As competitive pressure drives capex higher, the net economic surplus – the total payoff less investment costs – declines for the sector as a whole and could turn negative in adverse scenarios (Graph 11.B). Disappointment in returns could trigger a sudden pullback in financing and turn the capex boom into a protracted investment bust, with potential knock-on effects on financial conditions (see below).

Another risk is that the AI boom runs into a supply side roadblock. The AI build-out has recently been facing growing bottlenecks in electricity, advanced semiconductors and grid equipment. Fast-growing demand for computing power is already pressuring

electricity prices and input costs, with potential spillovers to inflation. Looking ahead, these temporary shortages may also amplify over-investment, as firms attempt to lock in future capacity through long-dated contracts that further expose them to any disappointments in demand.

Historical episodes of investment booms offer instructive parallels (Graph 11.C). The canal mania of the 1830s, the British railway mania in the 1840s, the electrification exuberance of the late 1920s (roaring 20s) and the dotcom boom of the late 90s all shared one common trait: a genuine technological breakthrough that attracted capital in excess of what commercial returns could ultimately justify. These episodes ended with an eventual reversal in investment, inducing economy-wide recessions. The scale and pace of the current AI investment boom accompanied by expectations of large productivity payoffs bear resemblance to these precedents, highlighting potential downside risks in the near term.

Financial vulnerabilities as amplifiers

Should inflation rise significantly or AI-led investment turn to a bust, the macroeconomic consequences could be amplified by existing financial vulnerabilities. A tightening of policy rates needed to contain inflation could precipitate a sharp pullback in asset prices after a prolonged period of exuberant risk-taking, triggering disruptive macro-financial feedback loops. A reversal of AI optimism could likewise have major financial consequences, given AI firms' rising leverage and growing footprint in credit markets. Vulnerabilities extend to their supplier ecosystem, including engineering, procurement and construction (EPC) contractors whose balance sheets are comparatively weak, leaving them exposed to any capex pullback by hyperscalers.

Equity valuations are elevated, particularly for firms at the core of AI development. The implied long-term earnings growth for the largest corporations sits well above recent historical benchmarks (Graph 12.A), with US stocks often trading at large premia to peers in other major markets. These implied rates often exceed even the elevated growth that some of the technology firms have delivered in their relatively short lifetimes. As these firms mature and command a larger share of the market, sustaining such high growth could become increasingly challenging.

Sentiment has also been a major driver of current valuations. Risk premia on the largest US stocks have compressed markedly since the Covid-19 pandemic, with the distribution shifting clearly to the left (Graph 12.B). This points to growing investor complacency and reduced compensation for risk-bearing. Post-pandemic exuberance has been largely broad-based across sectors and countries, coinciding with the rapid rise of AI as an investment theme following the release of generative AI tools in late 2022.

A major equity market correction could have larger macroeconomic consequences today than in the past. Household equity exposures have grown over the past few decades, both relative to total wealth (Graph 12.C) and income (Graph 3.C). A large correction in valuations could have more pronounced wealth effects and sharper consumption pullback than in the past. And with US stocks accounting for an outsized share of global equity markets – about 64% of the MSCI Global index – the wealth impact from a US-led repricing could propagate globally.

Financial stability could also be at risk in the event of an AI bust. Fixed income markets are one obvious vulnerability, given the high volumes of debt issued by hyperscalers, AI labs and EPC firms (Graph 13.A). Should hyperscalers slow or halt the aggressive pace of capex deployment, many borrowers across the supply chain could struggle to replace lost revenue and service their debt. The credit spreads of some AI firms have already begun to widen somewhat to reflect this risk (Graph 13.B), even as equity markets continue to price in significant upside gains

The opacity of AI-sector financing compounds these vulnerabilities. Hyperscalers, chip makers and AI labs are linked through a complex web of private arrangements. The most prominent is circular financing: chip makers and hyperscalers take equity stakes in AI labs or neocloud providers, who in turn commit to multi-year purchases of chips or computing power. Data centre construction is increasingly outsourced to third parties that lease facilities back to hyperscalers on long-dated contracts with embedded exit clauses. The terms of such deals are typically poorly

disclosed, with risks of the same asset being pledged multiple times. Together, such arrangements account for a sizeable share of sector-wide financing and forward revenue (Graph 13.C).

A sharp repricing of equity risk could prompt a reassessment of corporate credit risk and lead to tighter credit conditions more broadly.¹ Indeed, broad indices of credit spreads tend to correlate negatively with stock market returns (Graph 14.A), more so for the high-yield than the investment grade segment. While large, synchronised corrections in both markets are rare, there are notable precedents such as the Great Financial Crisis and the March 2020 dash for cash episode. A repricing of risk this time, whether triggered by higher interest rates or an AI bust, has the potential to be similarly disruptive by triggering a corporate credit freeze with wider implications for aggregate investment.

Any tightening in credit conditions could expose existing vulnerabilities in the less transparent private credit space, whose reach has expanded among middle market and small firms (Graph 14.B). Signs of stress are already visible: direct lending funds catering to retail investors have faced mounting redemption requests, forcing some to liquidate assets and return capital despite having no contractual obligation to do so. A larger shock, whether from a renewed inflation surge or a sharp AI-led repricing, could trigger a more widespread credit crunch. The consequences could extend beyond the non-bank perimeter, given banks' growing and opaque exposure to private credit funds, compounded by overlapping ties through insurance companies' balance sheets. Given that the affected segment is smaller corporates that account for a large share of job creation, the real economy implications could be substantial.

The growing role of private credit also raises concentration risks. Direct lending funds, dominant players in the private credit ecosystem, have quadrupled their lending to the AI and information technology (IT) sectors in the past five years, to about 15% of their portfolios.² These loans tend to be larger than those in other sectors, while their terms such as tenor and pricing remain broadly similar, raising questions about lending standards and risk pricing. Investor enthusiasm has allowed more funds to participate, increasing concentration risks as software firms draw on multiple private credit lenders simultaneously (Graph 14.C).

The banks and private credit lenders are concentrated in the SAME sectors, a collapse takes out a significant portion of their value and spreads across the entire US economy.

World Bank 15 [World Bank Group, "Banking Crisis", World Bank, 09-14-2015, <https://www.worldbank.org/en/publication/gfdr/gfdr-2016/background/banking-crisis>, accessed 7-25-2026] veer + mhl

Banking crisis

Banks are susceptible to a range of risks. These include credit risk (loans and others assets turn bad and ceasing to perform), liquidity risk (withdrawals exceed the available funds), and interest rate risk (rising interest rates reduce the value of bonds held by the bank, and force the bank to pay relatively more on its deposits than it receives on its loans).

Banking problems can often be traced to a decrease the value of banks' assets. A deterioration in asset values can occur, for example, due to a collapse in real estate prices or from an increased number of bankruptcies in the nonfinancial sector. Or, if a government stops paying its obligations, this can trigger a sharp decline in value of bonds held by banks in their portfolios. When asset values decrease substantially, a bank can end up with liabilities that are bigger than its

assets (meaning that the bank has negative capital, or is “insolvent”). Or, the bank can still have some capital, but less than a minimum required by regulations (this is sometimes called “technical insolvency”)

Bank problems can also be triggered or deepened if a bank faces too many liabilities coming due and does not have enough cash (or other assets that can be easily turned into cash) to satisfy those liabilities. This can happen, for example, if many depositors want to withdraw deposits at the same time (depositor run on the bank). It can also happen also if the bank’s borrowers want their money bank and the bank does not have enough cash on hand. The bank can become illiquid. It is important to note that illiquidity and insolvency are two different things. For example, a bank can be solvent but illiquid (that is, it can have enough capital but not enough liquidity on its hands). However, many times, insolvency and illiquidity come hand in hand. **When there is a major decline in asset values, depositors and other banks borrowers often start feeling uneasy and demand their money bank, deepening the bank’s troubles.**

A (systemic) banking crisis occurs when many banks in a country are in serious solvency or liquidity problems at the same time—either because they are all hit by the same outside shock or because failure in one bank or a group of banks spreads to other banks in the system. More specifically, a **systemic banking crisis is a situation when a country’s corporate and financial sectors experience a large number of defaults** and financial institutions and corporations face great difficulties repaying contracts on time. As a result, non-performing loans increase sharply and all or most of the aggregate banking system capital is **exhausted**. **This situation may be accompanied by depressed asset prices** (such as equity and real estate prices) on the heels of run-ups before the crisis, sharp increases in real interest rates, and a slowdown or reversal in capital flows. In some cases, the crisis is triggered by depositor runs on banks, though in most cases it is a general realization that systemically important financial institutions are in distress.

Systemic banking crises can be very damaging. They tend to lead affected economies into deep recessions and sharp current account reversals. Some crises turned out to be contagious, rapidly spreading to other countries with no apparent vulnerabilities. Among the many causes of banking crises have been unsustainable macroeconomic policies (including large current account deficits and unsustainable public debt), **excessive credit booms, large capital inflows, and balance sheet fragilities, combined with policy paralysis due to a variety of political and economic constraints.** In many banking crises, currency and maturity mismatches were a salient feature, while in others off-balance sheet operations of the banking sector were prominent.

A global database of banking crises was first compiled by Caprio and Klingebiel (1996). The latest version of the database, updated to reflect the recent global financial crisis, is available as Laeven and Valencia (2012). It identifies 147 systemic banking crises (of which 13 are borderline events) from 1970 to 2011. It also reports on 218 currency crises (defined as nominal depreciation of the currency vis-à-vis the U.S. dollar of at least 30 percent that is also at least 10 percentage points higher than the rate of depreciation in the year before) and 66 sovereign debt crises (defined by government defaulting on its debt to private creditors) over the same period. The database has detailed information about the policy responses to resolve crises in different countries. Analyses based on the dataset, such as Čihák and Schaeck (2010) suggest that consistently predicting banking crises is very difficult, but there are some variables (such as those capturing high leverage and rapid credit growth) that indicate increased likelihood of a crisis.

Chapter 2 of the Global Financial Development Report uses the Laeven and Valencia (2012) version of the database of banking crises to analyze what works (and what does not) in banking supervision and regulation. The chapter and the underlying paper (Čihák, Demirgüç-Kunt, Martínez Pería, and Mohseni-Cheraghloo 2012) use the responses from the World Bank’s Banking Regulation and Supervision Survey (accompanying the Global Financial Development Report) and performs an econometric analysis comparing countries that ended up in banking crises and those that managed to avoid them. The report and the paper find that crisis-hit countries had less stringent and more complex definitions of minimum capital, lower actual capital ratios, were less strict in the regulatory treatment of bad loans and loan losses, and faced fewer restrictions on non-bank activities. They had greater disclosure requirements but weaker incentives for the private sector to monitor banks’ risks. Overall, changes in regulation and supervision during the global financial crisis have been only gradual at best. Some changes, such as increasing capital ratios and strengthening resolution regimes, have gone in the right direction (making regulation in crisis countries closer to that in non-crisis countries), but at the same time, private sector incentives to monitor banks’ risks have been weakened by some of the policy interventions during the crisis. The analysis shows scope for strengthening regulation and supervision as well as private sector’s incentives to monitor risk-taking.

If companies cannot build in the United States, they will simply build somewhere friendlier.

Fayyad 26 [Abdallah, staff writer for the Boston Globe's Ideas section. He has worked at The Atlantic, Vox, and previously served on the Globe's editorial board from 2020 to 2023. He has appeared on CNN, MSNBC, NPR, the BBC, and other networks to discuss his work and the news, and guest hosted "Greater Boston" and "Talking Politics" on GBH; "The fight over data centers is a distraction"; 05/03/2026; The Boston Globe; <https://www.bostonglobe.com/2026/05/03/opinion/ai-data-center-moratorium/>; DOA: 08/06/2026] sumzom

But while opposing data center construction may make for good politics, it isn't moving the needle when it comes to regulating AI. It's merely a stand-in — a way to signal that you're pushing against a future that no one wants.

On an economic level, AI is expected to decimate the workforce. The CEO of Anthropic, one of the largest AI companies, has warned that AI could wipe out half of entry-level white-collar jobs. On a societal level, AI can be leveraged to promote misinformation and further undermine the public's trust in democratic institutions. And on the civilizational level, AI presents major cybersecurity risks and poses threats to human life, especially as it becomes increasingly integrated in military decisions and weaponry.

So there's a genuine interest in slowing down this technological advancement. And in the absence of robust federal regulations to fend off the worst possible outcomes, data centers make for a tangible target. They are starting to get more and more attention in elections across the country as candidates feel the need to address voters' concerns about them. In an interview with the Globe's editorial board, Graham Platner, a Democratic Senate candidate in Maine, floated the idea of regulating data centers like public utilities and potentially even making them publicly owned.

"There has never been a massive technological transformation that increases productivity that has ever done anything but disenfranchise and screw workers when left in the hands of corporate power," Platner said. "I actually think one of the best ways that we can do this is by having the infrastructure that supports it like data centers — I very much think we should have some form of public ownership of those. ... None of this runs without data centers."

That's partly why putting a moratorium on data center construction can be so appealing for politicians:

With the Trump administration's laissez-faire approach to AI, standing in the way of data center construction is one of the few routes states can take to tackle the broader regulatory problems around AI.

But if AI companies can't build data centers in the United States, they'll turn to friendlier jurisdictions elsewhere around the world, which would only compound Americans' privacy and data security concerns.

That doesn't mean that states should forget about putting guardrails around data centers. To the contrary, they should try to offset the potential social harms these facilities could impose. This regulation could come in various forms, including treating them like public utilities, taxing them in order to offset any potential increase in energy costs to consumers, and imposing requirements that would minimize their environmental risks. And it's especially important to put these sorts of regulations in place given that many states, despite all the local opposition, are competing to build data centers by offering tax breaks to companies instead of asking these multibillion-dollar corporations to pay their fair share when plugging into the grid.

But lawmakers at the local, state, and federal level shouldn't be under any illusions that regulating data centers, or blocking their construction altogether, could work as a temporary solution to the AI industry's unfettered growth. Ultimately, it's up to Congress to put up guardrails that will protect people from the risks posed by AI. And as long as the industry remains largely unregulated, nothing — not even a ban on the construction of new data centers — is stopping these corporations from abusing their power and using their technology in harmful and potentially dangerous ways.

History shows that moratoriums have removed hundreds of billions in investment into other regions. The affirmative will shift investment in the US to Nordic and Asian hubs.

Datacentres '26 [Datacentres provides real time market data, transaction tracking, and advisory services for the world's data centre infrastructure. 5-28-2026, "Europe's €156B Data Centre Freeze: How Policy Backlash Is,"

<https://www.datacentres.com/news/europe-s-156b-data-centre-freeze-how-policy-backlash-is-reshapin-g-global-investm-2026-05-28>, accessed 7-12-2026] baowow + kevin + mhl

Community opposition and **regulatory delays** have blocked or **stalled €156B in data centre projects** globally, **forcing operators to abandon traditional markets and accelerate shift to Nordic and Asian hubs**. The Policy Crisis Reshaping Global Data Centre Geography **The data centre industry faces an unprecedented policy headwind. Across Europe, North America, and Asia-Pacific, regulatory delays and community opposition have frozen approximately €156B in planned capacity** - equivalent to 18,200 MW of the current global pipeline. This represents a 73% increase in blocked projects compared to 2024, fundamentally **altering investment strategies and forcing hyperscalers to reconsider their geographic expansion plans**. The scale of disruption is stark. In **Ireland**, where Meta and Amazon have committed €8.2B in planned investments, Dublin City Council imposed a 12-month **moratorium on new data centre applications in February 2026**. **Netherlands** regulators extended permitting timelines to 24-36 months from the previous 12-month standard. **France** **tightened environmental impact assessments**, adding 8-12 months to project timelines. These are not marginal **delays** - they **represent existential challenges to the €280B hyperscaler**. **Cether policy incentives have** created a structural oversupply or simply **accelerated inevitable geographic rebalancing**. **For investors, the calculus is clear: Western European data centre development permits issued post-2025 now carry implicit valuations 18-24% higher due to policy risk**. The €156B in blocked capacity is a direct transfer of value to markets - Nordic, Asia-Pacific, and UK - offering permitting certainty. Policy, not technology or economics, now drives data centre geography.

Overseas infrastructure destroys governance AND invites sanctions. That chokes our economy.

Phillips-Robins 26 [Alasdair (Fellow, Technology and International Affairs Program), Teddy Tawil (Visiting Researcher, Technology and International Affairs Program), Sam Winter-Levy (Senior Fellow, Technology and International Affairs), 06-08-2026, "The Compute Coalition: How to Build the Future of AI in the Free World", Carnegie Endowment for International Peace, <https://carnegieendowment.org/research/2026/06/the-compute-coalition-how-to-build-the-future-of-ai-in-the-free-world#ii-the-geopolitics-of-compute>, accessed 07-15-2026] wrong + mhl

Military and intelligence. **A country that can train and run defense-relevant AI models on home soil retains control over the models, the data they process, and the operational security of the entire pipeline. Without domestic compute, a state's military and intelligence capabilities could become dependent on foreign providers**. In a crisis, **access might be throttled or cut when it is needed most**.^{29¶} Governance. **States with significant domestic compute are far better positioned to shape international standards on AI safety, military use, and other dimensions of governance than ones that depend entirely on others for access to the underlying infrastructure**. While the countries that are home to the most advanced model developers have the first claim on setting the rules, **those that provide the infrastructure on which the model is trained and run will also have a substantial say over what safety evaluations are required**, what values are embedded, and what uses are permitted or prohibited.[¶] Resilience. **Dependence on foreign compute creates vulnerabilities** beyond the military domain. **AI capabilities cannot be stockpiled**, and **countries that rely on remote providers face the possibility that**

access could be restricted—whether because exporters want to squeeze prices or redirect limited computing capacity toward other priorities, or as a result of sanctions, export controls, corporate policy changes, or external physical or cyberattacks. As one analyst has put it, “Being a net importer of a core commodity in the modern economy is seldom an excellent place to be.”¶ Fiscal stability. If AI-driven automation shifts a substantial share of economic output from labor income to corporate income, countries that host none of the relevant corporate activity may see their tax base erode with no corresponding new revenue stream. Locating as much frontier AI activity as possible within a country’s tax jurisdiction—through data centers, local subsidiaries, or negotiated revenue-sharing arrangements—may provide a pathway to fiscal stability that pure importers of AI services will lack.¶ A country or coordinated bloc that hosts a meaningful share of global AI infrastructure could gain something analogous to what OPEC members hold in energy markets.¶ A country or coordinated bloc that hosts a meaningful share of global AI infrastructure could also gain something analogous to what OPEC members hold in energy markets. Even a single-digit share of global supply, if it can credibly be taken offline, would likely be enough to inflict real costs. Onshore compute does not guarantee sovereignty over AI. But it does make it far more likely that a country can influence how AI develops.

Without sustained economic growth an economic crash would be devastating. The recession of 2008 proves that a US-led decline causes a global economic crash and puts 100 million into poverty. The AI recession would be worse because the whole economy would be focused on one sector that would tank all at once.

Harris ‘14 [Jonathan M. Harris is Senior Research Associate at the Tufts University Global Development and Environment Institute. His current research is on the macroeconomics of global environmental issues, particularly climate change. Ph.D., Boston University 1988; M.A., Political Economy, Boston University, 1981; B.A. cum laude, Economics, Harvard College, 1971. January 2014. “The Financial Crisis and the Great Recession”. In book: Macroeconomics in Context (pp.337-356), Edition: 2, Chapter: 15, Publisher: M.E. SharpeBoston. Tufts University. DOA 8-28-2026]
https://www.bu.edu/eci/files/2019/06/MAC_2e_Chapter_15.pdf mhl

The financial crisis that commenced in 2007 and its aftermath have been widely referred to as the “Great Recession”—and with good reason. From its beginning until its nadir in 2009, it was responsible for the destruction of nearly \$20 trillion worth of financial assets owned by U.S. households. During this time, the U.S. unemployment rate rose from 4.7 percent to 10 percent (not counting the discouraged and marginally attached workers discussed in Chapter 7). By 2010, college graduates fortunate enough to find a job were, on average, earning 17.5 percent less than their counterparts before the crisis—and experts were predicting that such a decline in earnings would persist for more than a decade. The crisis also spread beyond U.S. borders. As consumption and income declined in the United States, many countries experienced a significant reduction in exports as well as a decline in the investments that they held in the United States. As a result, global GDP declined by 2 percent in 2009. It has been estimated that between 50 million and 100 million people around the world either fell into, or were prevented from escaping, extreme poverty due to the crisis. Why did this happen? Why were its effects so long-lasting? What lessons can be learned for the future? These are complicated questions to which this chapter provides some answers.

constructive end

On their contention about potential grid collapse

Alt causes to grid collapse.

Mulligan 26 [Grant Mulligan is a writer, investor, conservationist, and former Roots of Progress Institute Fellow. 4-22-2026. "The power grid is breaking. Can it fix itself?". Big Think. DOA: 7/30/26. <https://bigthink.com/science-tech/the-power-grid-is-breaking-can-it-fix-itself/>] tbt

The electric grid can fail for multiple reasons, but the most common is physical damage. Imagine a power pole that snaps in the wind. Risks to physical infrastructure include basic equipment failures, damage from wildlife, and extreme weather events, which are becoming more common due to climate change.

Bad actors can also attack physical infrastructure — Russia has launched physical attacks on Ukraine's energy grid in an attempt to cripple the economy and crush morale. They can also target the digital infrastructure that keeps everything working. According to industry data cited in a report by the International Energy Agency, a typical utility experienced more than 1,500 cyberattacks per week in 2024.

HDC aren't key to putting strain on the grid, there is no grid collapse.

Prefer 23 years of analysis.

CFR 24 [Council on Foreign Relations (CFR) Education is an initiative of the Council on Foreign Relations, a nonpartisan think tank and publisher. 12-4-2024. "Why Electric Grids Need to Be More Resilient". CFR Education. DOA: 7/30/26. <https://www.cfr.org/education/learn/reading/grid-resiliency-climate-change>] tbt

The northeastern United States is not the only place where electricity systems are vulnerable to extreme weather. And electricity infrastructure doesn't need a hurricane to shut down.

The nonprofit Climate Central estimates that roughly 80 percent of U.S. power outages in the twenty-first century have been caused by weather. Over half those outages were caused by severe weather, including high winds and intense storms.

Climate change makes those weather events worse—not just in the United States. Events like heat waves, tropical storms, flooding, wildfires, and even winter storms are increasing in intensity, frequency, or both. Such weather can damage energy infrastructure and make power transmission less efficient. They can also drive increased demand for power; during heat waves, for example, people use more energy to keep their homes cool. Increased use can cause power outages, often at moments when people need electricity the most. Those outages are damaging people and economies. And without updates to make energy grids more resilient, they could keep getting worse.

As of RIGHT NOW, Federal regulations are ALREADY solving now without a moratorium.

Sorg 26 [Lisa Sorg is the North Carolina reporter for Inside Climate News. A journalist for 30 years, Sorg covers energy, climate environment and agriculture, as well as the social justice impacts of pollution and corporate malfeasance, Charles Paullin is a Richmond, Virginia-based reporter focusing on energy and environment issues, 6-18-2026, "Federal Regulators Tell Electric Grid Operators to Fix Their Rules on Data

Centers," Inside Climate News,
<https://insideclimatenews.org/news/18062026/federal-energy-regulatory-commission-data-center-orders/>, DOA: 7-31-2026] emman

Today federal energy regulators issued sweeping orders that require the nation's six major grid operators within 60 days to propose reforms or justify their rules that govern how data centers and other large customers connect to the electric grid.

Additionally, within 30 days, each grid operator and its transmission owners must submit a detailed report describing how the grid operator will ensure there is adequate generation to serve existing and new large loads.

The purpose of the order is to expedite connections between large customers and utilities, but with some proposed protections for residential and small commercial customers.

The wholesale energy market for two-thirds of the United States, roughly 200 million customers, is managed and monitored by regional grid operators, such as PJM Interconnection, Midcontinent Independent System Operator (MISO) and the Electric Reliability Council of Texas (ERCOT).

U.S. Secretary of Energy Chris Wright ordered the Federal Energy Regulatory Commission (FERC) last October to begin rulemaking to address energy demands and grid stability related to large customers, in particular data centers for AI and cryptomining.

Although FERC's order accounts for regional differences, its common goal is to stabilize the grid, add transparency to the ratemaking process and prevent utilities from shifting additional data-center energy costs to residential ratepayers.

"This product is not as ambitious as what Secretary Wright asked them to do," said Nick Guidi, a senior attorney for the Southern Environmental Law Center. "I think they responded in a way that makes sense for traditional FERC activity, in a way that seems like it has some pluses and minuses."

The order also requires grid operators to examine how they accommodate co-location agreements, which allow data centers to be built at or near power plants, and "behind-the-meter" energy supplies, in which the data centers themselves build their own power plants.

Additionally, within 30 days, grid operators must submit a detailed report describing how they intend to ensure that adequate generation will be available to serve existing and new large loads.

The order allows grid operators to define "large load." Many states have set a threshold of 100 megawatts, but as more of these customers connect to the grid, it could be lowered.

Grid repairs require funding that ONLY hyper-scale data centers can provide. Preserve econ to preserve grid

Hedger 26 [Patrick Hedger is the Director of Policy at NetChoice, 07-09-2026, "New Study Confirms What We've Been Saying: Data Centers Are Lowering Your Electric Bill," NetChoice, <https://netchoice.org/new-study-confirms-what-weve-been-saying-data-centers-are-lowering-your-electric-bill/>, DOA: 07-28-2026] emman

For the past year, data centers have become a convenient scapegoat for rising utility bills. State legislatures have even floated construction bans, and local governments have passed moratoriums as a result. The uproar led to the White House intervening in March 2026 with the "Ratepayer Protection Pledge," an assurance that data centers would not raise consumer electric bills. The economic evidence backs this assurance.

The Conventional Wisdom — And Why It's Wrong

A new working paper from researchers at EPRI and Watershed set out to answer a simple question with real data instead of assumptions: did the recent boom in data center construction actually raise what Americans pay for electricity? Using an instrumental variables approach, a statistical method designed to isolate cause from mere correlation, the researchers found that data centers actually caused average U.S. residential electricity rates to fall, not rise, between 2015-2024. A doubling of a state's data center capacity was associated with a roughly 3.5% decrease in residential rates, holding household electricity use constant.

How did more demand lead to lower prices? The answer lies in basic power-sector economics that gets lost in the current political noise. Electricity retail rates aren't set purely by marginal supply and demand the way, say, gasoline prices are. The grid carries enormous fixed costs due to the poles, wires, substations, and power plants that exist and operate regardless of how much electricity flows through them. Because of this, they operate using a rate structure that recovers those costs across every kilowatt-hour sold. When durable new demand like a data center comes online, it spreads those fixed costs over more usage, and it often justifies newer, more efficient generation that costs less per unit than the aging assets already on the books. The net effect, as the study's authors describe it, is that new demand can lower average costs for everyone in the system.

This aligns with what NetChoice has argued for months: large data centers shoulder a growing share of utilities' fixed costs, which reduces demand charges on residential bills. The real culprits behind recent rate increases include aging grid infrastructure, extreme weather disasters, state renewable mandates, and the retiring of natural gas, coal, and nuclear power plants — not the servers powering America's digital economy.

A Case Study: Virginia v. California

Virginia is the nation's data center capital, hosting more capacity than any other state, and where data centers draw over 20% of the state's electricity. If the "data centers spike your bill" narrative were true, Virginia's ratepayers should be feeling it acutely. Instead, the study finds Virginia's residential rate increases tracked closely with the national average, with no meaningful outlier effect occurring from being the epicenter of American data center growth. Virginia's experience is a live, large-scale test of the theory, and the results undercut the fear-based narrative driving opposition in statehouses around the country.

Contrast that with California, which saw the single largest residential electricity price increase of any state in the study period, nearly 40%, despite having comparatively modest data center growth relative to its size. The driver wasn't AI infrastructure; it was wildfire-related grid hardening costs layered onto the state's utilities. California is the clearest available evidence that when electricity bills spike, the explanation usually lies elsewhere, not in the presence of a data center down the road.

What This Means for the Policy Debate

Put the two case studies together with the paper's national-level finding, and a consistent picture emerges: states adding the most data center capacity have not been the states seeing the steepest electric bills. If anything, the correlation runs the other way. The study also specifically tested and rejected the fear that data centers are quietly shifting costs onto residential customers while enjoying discounted industrial rates.

The efficiency gains from new demand appear to flow to households, not away from them.

The researchers are careful to note real caveats, like turbine backlogs, transformer shortages, tariffs on solar equipment, and stalled permitting for new generation, which could eventually make new capacity more expensive to build, reversing the price-lowering effect if supply can't keep pace with demand. That's precisely why NetChoice has consistently called for smart, competitive state policy: modern permitting processes, interconnection reform, and tax codes built for large capital investment, rather than blanket bans that would only choke off the very grid investment the data shows is good for ratepayers.

Data centers are the backbone of America's digital infrastructure and economic activity, and now there is rigorous economic evidence to back up the fact that growing, durable electricity demand tends to make the whole system cheaper to run, not more expensive. As more states debate legislation this year that hinders the data center buildout in America, policymakers should look past the fear-driven talking points and toward the data. The next time a headline blames your electric bill on the data center down the street, the receipts probably say otherwise. Lawmakers should let facts, not fear, drive decisions on the infrastructure powering America's economic future.

Independently, self-healing capabilities and AI efficiency guarantees grid reliability--but is dependent on 'computing capabilities.' which is exactly what HDC's provide.

Mulligan 26 [Grant Mulligan is a writer, investor, conservationist, and former Roots of Progress Institute Fellow. 4-22-2026. "The power grid is breaking. Can it fix itself?". Big Think. DOA: 7/30/26. <https://bigthink.com/science-tech/the-power-grid-is-breaking-can-it-fix-itself/>] tbt + emman

"A **self-healing grid** is a power system that **proactively detects disturbances, isolates the problem, and restores service automatically, often in seconds.**" says Massoud Amin, the "father of the smart grid" and professor emeritus at the University of Minnesota. "**It uses sensors, communications, and advanced controls distributed across the network to monitor the system and take corrective action continuously.**" The goal is simple: prevent small disturbances from turning into large outages."

A self-healing grid is not an endpoint. **It's a spectrum of capabilities.** Together, they transform the grid from a **passive network into an adaptive system** that enables utilities to "identify and respond to issues and outages faster and mitigate them in as close to real-time as possible," says Flemming.

For example, **while a self-healing grid can't automatically repair physical damage** — if the wind snaps a power pole, a lineman still needs to go out and repair it — it can detect the problem almost immediately, **automatically open and close switches across the network to route power away from the downed line, and stabilize the grid to prevent cascading failures and minimize the area affected by outages.** No need to wait for customers to report the downed line.

This is fundamentally different from how the grid has historically operated. Instead of being a static, centrally managed system, **the grid is becoming a dynamic, data-driven one that can respond to conditions in real time.**

"The traditional grid is largely reactive," says Amin. "When something fails, equipment trips, and crews respond after the outage occurs. A self-healing grid is aware and adaptive. It senses disturbances in real time, isolates damaged sections, and reroutes power automatically. The difference is speed and intelligence."

Self-healing capabilities like the above — **automated switches that can quickly isolate faults and reroute power** — are already being deployed on local distribution grids. Distribution infrastructure is prioritized because it makes up the vast majority of the grid and accounts for most outages. **Duke Energy Florida reported that investments in self-healing technologies helped avoid 280,000 extended power outages** and more than 300,000 hours of outages for customers in 2025.

Other **technological advances** are helping modernize the grid by allowing operators to determine the **real-time capacity limits of transmission lines** — how much electricity they can carry before conductors overheat. Historically, these limits have been static, conservative, and not entirely accurate — how much power a transmission line can actually carry safely varies depending on weather conditions.

In March, **PJM Interconnection** became **the first grid operator to comply with a Federal Energy Regulatory Commission order to adjust transmission line capacity ratings using real-time weather data.** PJM now uses sensors and software to collect **and analyze weather conditions and forecasts,** allowing it to update **transmission limits hourly instead of relying on a fixed rating.** Research from the DOE's Idaho National Laboratory suggests that implementing dynamic ratings could increase usable transmission capacity by roughly 10% to 40%.

AI also has a role to play in grid modernization. In 2024, researchers at NREL published a **report describing "eGridGPT," a generative AI model "engineered to virtually support power grid control room operators by assisting in decision-making processes and interpreting the data and models."** PJM is already using AI for interconnection planning to accelerate the connection of new generation resources to the grid, while NextEra Energy and Google Cloud recently

announced a collaboration to deploy AI to enhance both field operations and grid management and optimization efforts.

However, AI is not a panacea. I asked several experts about AI's potential role in achieving self-healing grids, and they were united in their belief that AI would not be the ultimate operator of the grid. "[AI] expands the operator's field of vision," says Amin, but it's still up to the human to make the important decisions. In many respects, AI's real advantage to the grid operator comes down to speed — by quickly analyzing the troves of data collected by sensors across a modern grid, it can help operators make decisions faster.

Ultimately, it'll be a combination of better sensors and computing capabilities that will meaningfully improve the capacity and resilience of the grid. As these technologies advance, utilities will be able to deploy them on a wider scale and in more contexts — making the grid more resilient for more people.

"Modern society runs on electricity," says Amin. "Our security, economy, and quality of life all fundamentally depend on reliable, affordable, and sustainable electricity. When the grid fails, everything else begins to fail with it."

AI models are key to efficiency.

Mulligan 26 [Grant Mulligan is a writer, investor, conservationist, and former Roots of Progress Institute Fellow. 4-22-2026. "The power grid is breaking. Can it fix itself?". Big Think. DOA: 7/30/26. <https://bigthink.com/science-tech/the-power-grid-is-breaking-can-it-fix-itself/>] tbt

AI also has a role to play in grid modernization. In 2024, researchers at NREL published a report describing "eGridGPT," a generative AI model "engineered to virtually support power grid control room operators by assisting in decision-making processes and interpreting the data and models." PJM is already using AI for interconnection planning to accelerate the connection of new generation resources to the grid, while NextEra Energy and Google Cloud recently announced a collaboration to deploy AI to enhance both field operations and grid management and optimization efforts.

However, AI is not a panacea. I asked several experts about AI's potential role in achieving self-healing grids, and they were united in their belief that AI would not be the ultimate operator of the grid. "[AI] expands the operator's field of vision," says Amin, but it's still up to the human to make the important decisions. In many respects, AI's real advantage to the grid operator comes down to speed — by quickly analyzing the troves of data collected by sensors across a modern grid, it can help operators make decisions faster.

Ultimately, it'll be a combination of better sensors and computing capabilities that will meaningfully improve the capacity and resilience of the grid. As these technologies advance, utilities will be able to deploy them on a wider scale and in more contexts — making the grid more resilient for more people.

"Modern society runs on electricity," says Amin. "Our security, economy, and quality of life all fundamentally depend on reliable, affordable, and sustainable electricity. When the grid fails, everything else begins to fail with it."

Additionally collapses would be compartmented. Theoretical US grid the aff states will not collapse the whole US.

Einberger 23 [Mathias Einberger is a Manager for the Carbon-Free Electricity Program at RMI. 1-12-2023. "Reality Check: The United States Has the Only Major Power Grid without a Plan". RMI. DOA: 7/30/26. <https://rmi.org/resources/the-united-states-has-the-only-major-power-grid-without-a-plan/>] tbt

The US power grid is, in fact, highly fragmented and consists of not one, but three different sections. These are called the Eastern, Western, and ERCOT interconnections — three separate power grids that are almost completely isolated from one another, electrically speaking. To make matters worse, the high-voltage, long-distance electric transmission lines that form the backbone of each of these grids **are largely planned in even greater local isolation.**